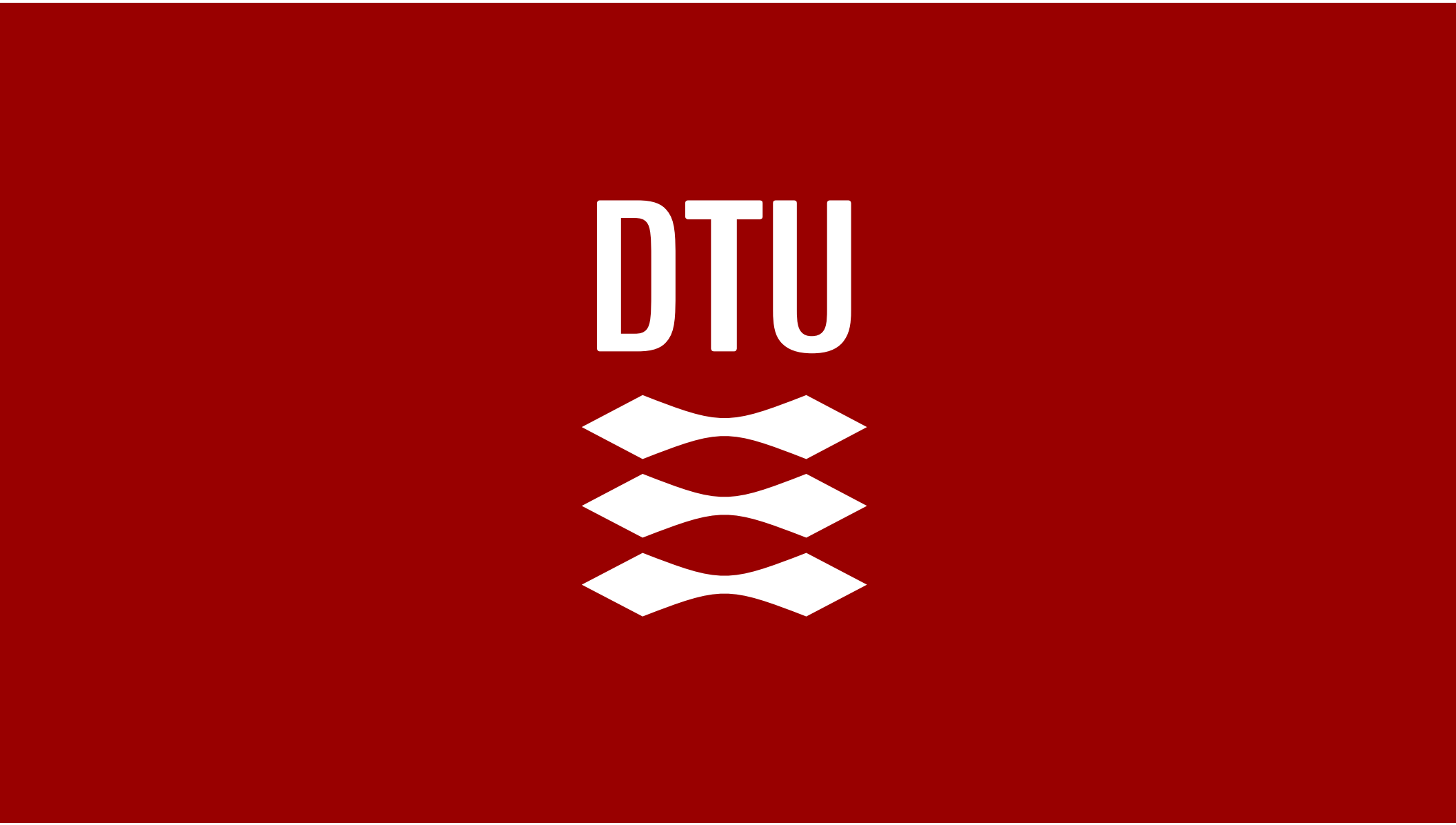
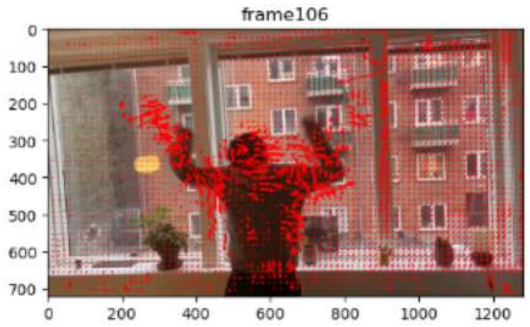
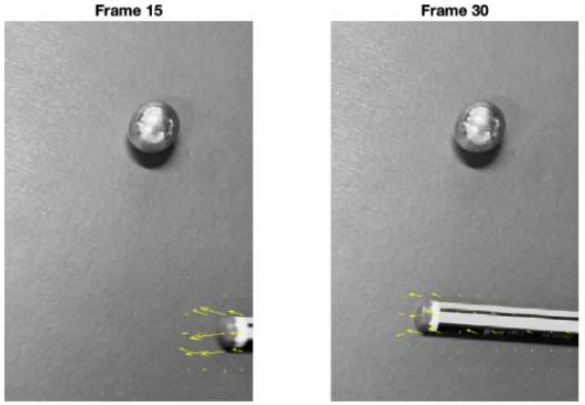

DTU

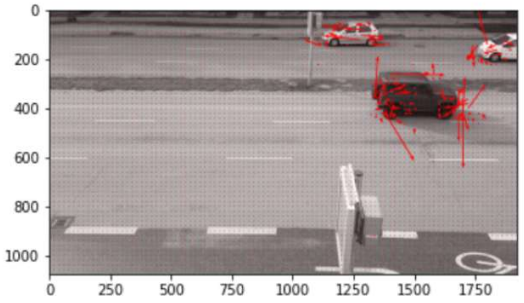
Videos

- **Categorization I**
  - Human/sport*[1]
  - Vehicle*
  - Simple object (spin, roll, swing)
  - Animation
  - Material interaction
  - Camera stabilization

- **Categorization II**
  - Controlled/scientific
  - Goofy/sporadic
  - …

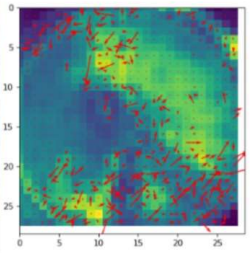








(e) Oil problem original



(f) Oil problem result

Considerations for selecting e.g. BSc/MSc projects:

Point 1: Experimentation
- Yes (Practical)
- No (Theoretical)

Point 2: Data
- Serious (Result)
- "Random" (Model)

*[1]TrackMan

Introduction (Discussion & Conclusion)

1. Entice the reader
2. Onset for the ”story telling”
   – Data, Discussion and Conclusion

- **General remarks:**
  – Lack of purpose
  – Strong indications of it being an exercise

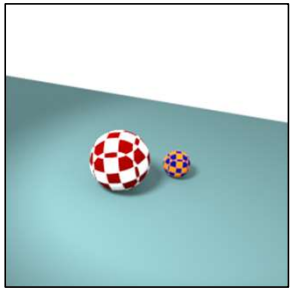
Considerations for selecting e.g. BSc/MSc projects:

Point 3: Introduction and purpose
– Fixed (little freedom)
– Flexible (lots of freedom)

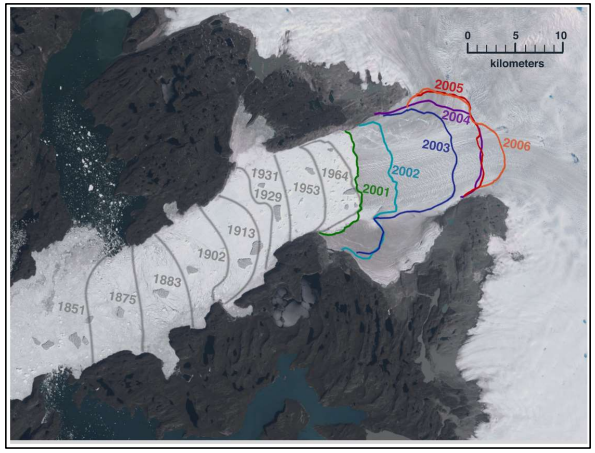
1 Theory & Background ≠ Introduction

Optic flow is the pattern of apparent motion of objects, surfaces and edges in a video caused by the relative motion between the observer and the environment. To say it bluntly, Optic flow is a method to estimate the motion in a sequences of images (video). Some of it's benefits is that optic flow can be used for frame interpolation. If your video of your object is moving very fast and you want to apply slow motion, then the video might become laggy. Optic flow could then be used to predict the interpolating missing frames and therefore then smooth the image. Other uses include image stabilization where drones or other similar objects require that the video is optimally stable. Optic flow was invented in the 1940's and is still used in technologies today.

Toy problem:



Glacier recession:



https://svs.gsfc.nasa.gov/3395

Introduction

- Introduction/purpose is not fixed, although the data is!



Concrete exercise for the next report:

End* your introduction with a bullet point* stating:

- **What is the purpose/goal of the report?**

Example 1:

– Purpose: ~ investigate **if** **Mx** is suitable for the application/problem

– Discussion: Sufficiently "accurate", "generalizable", "scalable" and "tunable" etc.?

– Conclusion: "No!" or "probably not" or "potentially" or "Yes!"

Example 2:

– Purpose: ~ **compare** the potential of **M1**$_{(simple)}$* vs. **M2** $_{(advanced)}$* for the application/problem

– Discussion: Compare "accuracy", "generalizability", "scalability" and "tunability" etc.

– Conclusion: **M1** or **M2** or "depends on *x*"

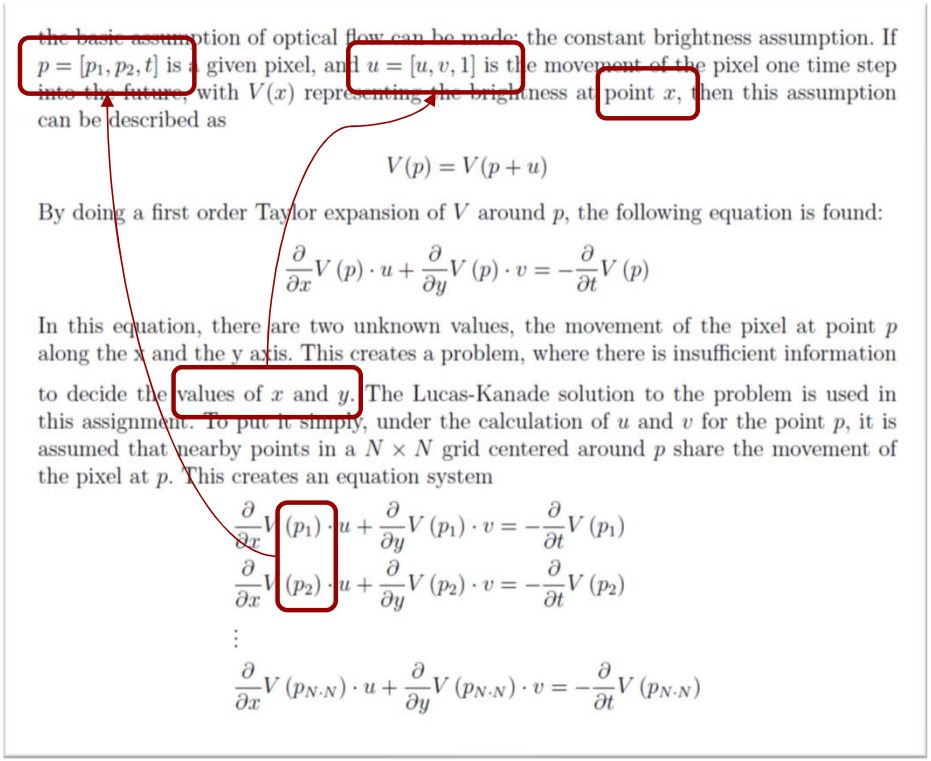
*It doesn't have to be the actual end nor an actual bullet point

Math & Methodology

- **General remarks:**

1. References
   - Can you cite the exercise text?

2. Notation formatting and consistency
   - Follow conventions wrt. bold and italic fonts, etc.
     - **Stringent** or **pedantic** supervisors/reviewers*.
     - Get familiar with it now! Gain experience with notation, procedures, TEX packages etc.



the basic assumption of optical flow can be made the constant brightness assumption. If $p = [p_1, p_2, t]$ is a given pixel, and $u = [u, v, 1]$ is the movement of the pixel one time step into the future, with $V(x)$ representing the brightness at point x, then this assumption can be described as

$$V(p) = V(p + u)$$

By doing a first order Taylor expansion of V around p, the following equation is found:

$$\frac{\partial}{\partial x}V(p) \cdot u + \frac{\partial}{\partial y}V(p) \cdot v = -\frac{\partial}{\partial t}V(p)$$

In this equation, there are two unknown values, the movement of the pixel at point p along the x and the y axis. This creates a problem, where there is insufficient information to decide the values of x and y. The Lucas-Kanade solution to the problem is used in this assignment. To put it simply, under the calculation of u and v for the point p, it is assumed that nearby points in a $N \times N$ grid centered around p share the movement of the pixel at p. This creates an equation system

$$\frac{\partial}{\partial x}V(p_1) \cdot u + \frac{\partial}{\partial y}V(p_1) \cdot v = -\frac{\partial}{\partial t}V(p_1)$$
$$\frac{\partial}{\partial x}V(p_2) \cdot u + \frac{\partial}{\partial y}V(p_2) \cdot v = -\frac{\partial}{\partial t}V(p_2)$$
$$\vdots$$
$$\frac{\partial}{\partial x}V(p_{N \cdot N}) \cdot u + \frac{\partial}{\partial y}V(p_{N \cdot N}) \cdot v = -\frac{\partial}{\partial t}V(p_{N \cdot N})$$

Useful TEX command:

```
\newcommand{\vect}[1]{\ensuremath{\mathbf{#1}}}
```

*[Cough]: mathematicians

Results & Visualization [3 seconds]:

- How fast can you ‘understand’ this figure?

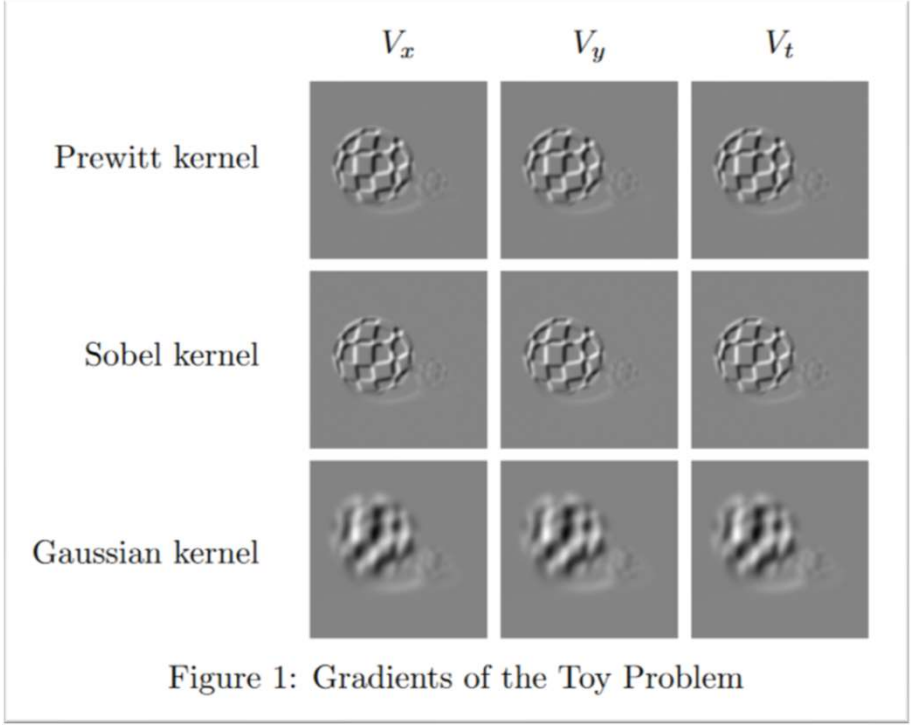


Figure 1: Gradients of the Toy Problem

Results & Visualization [Discuss 3 minutes]

- Discuss [context: the report text read nicely]:
  - Does the cropout work?
  - Something missing about the text?
  - Is the caption sufficient?
  - Does the grid layout work?
  - Anything else to comment?



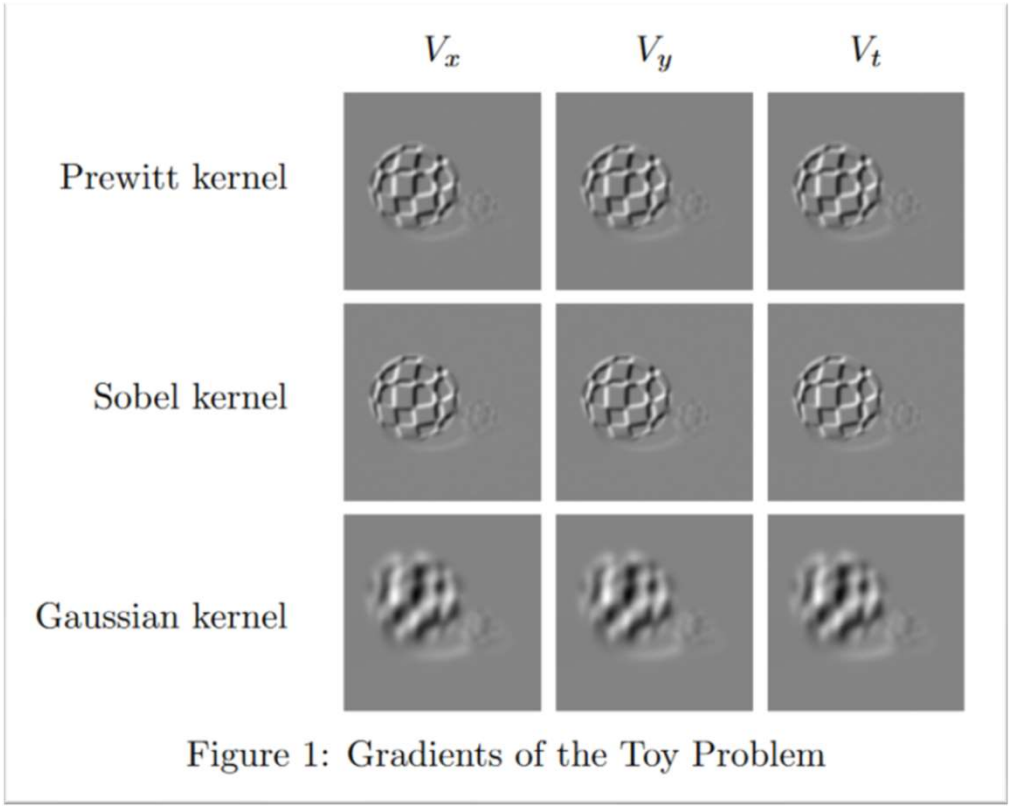


Figure 1: Gradients of the Toy Problem

Results & Visualization:

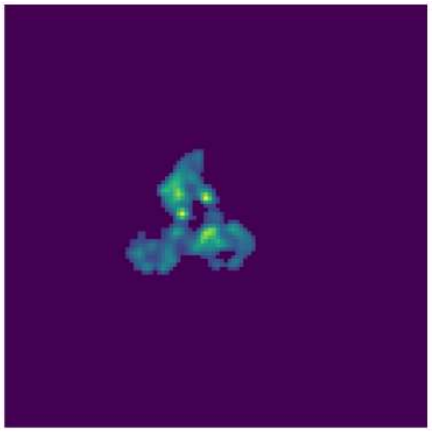
- **General remarks:**
  1. Embedded videos in pdf!
  – However nice they are, be aware of:
     - Communication to the reader
     - Translation to print
     - Only (?) works in Adobe

  2. Be aware of: ”Colormaps”, ”Colorbars”, ”Background color”, ”Axis scaling”, ”Axis ticks and labels”.

  3. *”A picture is worth a thousand words”* : A thousand words takes effort to write ‘perfectly’!

Vector magnitude as an image!
(colorbar needed perhaps)





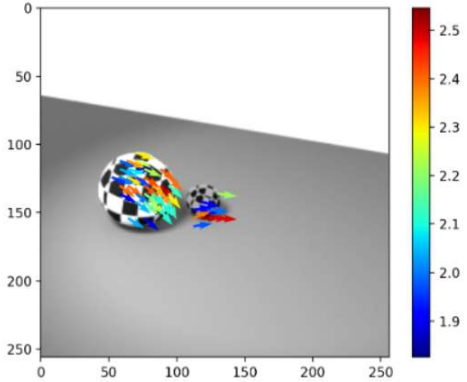
Results on the original RGB image!

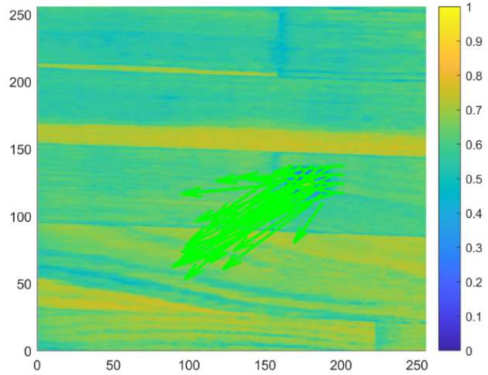
Orientation coloring!
(remember colorbar/-colorwheel)



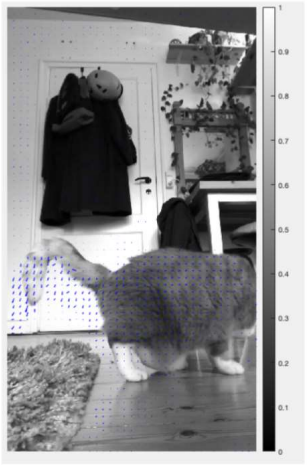
Results & Visualization:

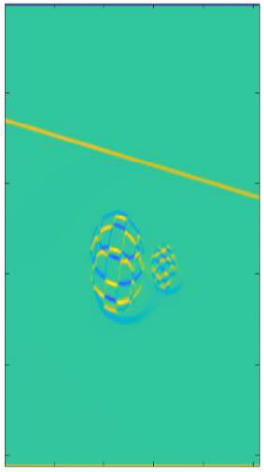
1. **Colormaps**
   - *Gray* for scene, √
   - *Parula* for scene, ?
   - *Jet* for magnitude, ?
   - *Parula* for gradient, ?





2. **Colorbars**
   - For scene, %
   - For magnitude, √ (remember label!)





3. **Axis scaling**
4. **Axis ticks + tick labels**
5. **Background color**
6. **Frame box**
7. **Vectors:**
   - Density and size
   - Is color the most intuitive representation of magnitude?

Focus points

- **Linking:** Introduction to Discussion to Conclusion
  - Ensure a goal for the report*
- Consistent **math notation**
- **Visualization:**
  - Colormaps & colorbars
  - Axis scaling, labels and ticks

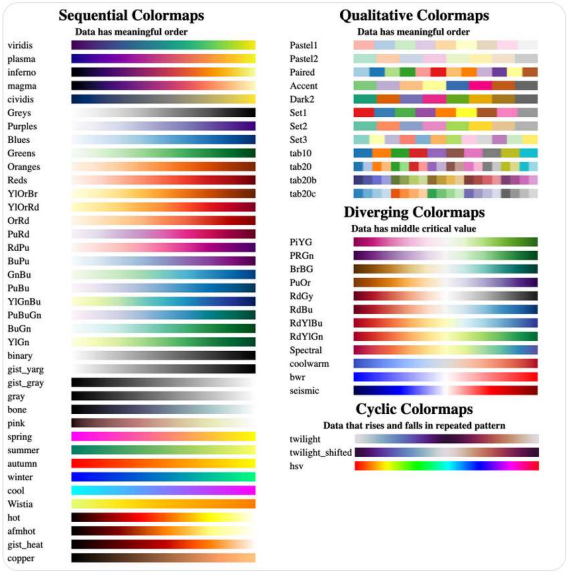
Tweet

Teddy Petrou
@TedPetrou

Matplotlib colormaps - 4 major categories for different types of data

Sequential - data has meaningful order
Qualitative - data has no meaningful order
Diverging - data has middle critical value
Cyclic - data rises and falls in repeated pattern

buff.ly/2UDnej5



https://twitter.com/tedpetrou/status/1246110879118176257

*Which is not [paraphrasing] "*We solve an exercise, because we were told to do so*"